EXPERIMENT: DESIGN OF A RECTANGULAR MICROSTRIP PATCH ANTENNA USING ANSYS HFSS

1. Creating a New Project

1. Open ANSYS HFSS.
2. Navigate to Project Insert HFSS Design.

2. Creating the Ground Plane

1. Select Create Rectangle and rename Rectangle1 as Ground in the Model Window.
2. Set the rectangle properties:

- Size (X × Y): 60 × 60 mm
- Centre: (-30, -30, 0) mm
- Change the colour and click Apply OK.

3. Creating the Substrate

1. Select Create Box and draw a box above the Ground plane with the same X and Y dimensions.
2. Rename Box1 as Substrate in the Model Window.
3. Set the properties:

- Material: FR4_Epoxy
- Size: 60 × 60 × 1.6 mm
- Z-height: 1.6 mm
- Change the colour and click Apply OK.

4. Creating the Patch

1. Select Create Rectangle and rename Rectangle1 as Patch.

2. Set the rectangle properties:

- Length (L): 29.4 mm
- Width (W): 38 mm
- Centre: (-29.4/2, -38/2, 1.6) mm
- Change the colour and click Apply OK.

5. Creating the Feedline

1. Select Create Rectangle and draw a rectangle along the X-axis.
2. Rename Rectangle1 as Feed.
3. Set the properties:

- Size (X × Y): 30 × 3 mm
- Centre: (0, 3/2, 1.6) mm
- Change the colour and click Apply OK.

6. Creating the Cut

1. Draw a rectangle over the feed region and rename it as Cut.
2. Set the properties:

- Size: (-9.5, 5) mm
- Position: (29.4/2, -5/2, 1.6) mm.

3. Select Patch and Cut together.
4. Click Subtract to create the cut.

7. Connecting the Feedline and Patch

1. Select Patch and Feed together.
2. Click Unite to combine both structures into a single conducting element.

8. Assigning the Excitation Port

1. Under the Visibility tab, change the working plane to YZ.
2. Select Create Rectangle and draw a rectangle on the substrate.
3. Rename it as Port and set its Y-dimension to 3 mm.
4. Select the Port and navigate to Right-click Assign Excitation Lumped Port.

5. Change the Integration Line to New Line and draw a line through the centre of the rectangle.
6. Verify that the Integration Line status changes to Defined.
7. Click Next Finish and verify the port under the Project Manager Window.

9. Creating the Radiation Boundary

1. Select Create Box and draw a box surrounding the entire antenna structure.
2. Rename the box as radiation.
3. Set the following properties:

- Medium: Air
- Size (X × Y × Z): 80 × 80 × 40 mm
- Centre/Position: (-40, -40, -20) mm
- Transparency: 1

4. Apply the settings and click OK.

10. Assigning Boundary Conditions

1. Select Patch and Ground together.
2. Right-click and navigate to Assign Boundary—Perfect E.
3. In the Model Window, right-click Rad1 and select Assign Boundary—Radiation.

Results

11. Setting Up the Simulation

1. Navigate to HFSS —Analysis Setup —Add Solution Setup—Advanced.
2. Configure the solution parameters:

- Solution Frequency: 2.4 GHz
- Maximum Number of Passes: 20

3. Configure the frequency sweep:
   Parameter Value
   Sweep Type Fast
   Distribution Linear Step
   Start Frequency 1 GHz

Stop Frequency 3 GHz
Step Size 0.02 GHz

4. Click OK to save the settings.

12. Validation and Simulation

1. Navigate to HFSS → Validation Check to verify the design.
2. If validation is successful, select HFSS → Analyze All.
3. Wait for the simulation to complete.

13. Plotting the S-Parameters

1. In the Project Manager Window, right-click Results.
2. Select Create Terminal Solution Data Report → Rectangular Plot.
3. Set the following parameters:

- Category: Terminal S Parameter
- Function: dB

4. Click New Report to generate the S-parameter plot.
5. Right-click the plot and select Marker → Add Marker.
6. Place the marker at the dip in the curve to identify the resonant frequency.

14. Creating the Far-Field Setup

1. Navigate to HFSS → Radiation → Insert Far Field Setup → Infinite Sphere.

2. Set the angular ranges:
Parameter Start Stop
Phi (φ) 0° to 360°
Theta (θ) 0° to 180°

3. Apply the settings.

15. Plotting the Radiation Pattern

1. Navigate to Results → Create Far Field Report → 3D Polar Plot.
2. Select rE as the quantity and dB as the function.
3. Click New Report to generate the radiation pattern.

16. Visualizing the Radiation Pattern on the Structure

1. Navigate to HFSS → Fields → Plot Fields → Radiation Field.
2. In the window, enable the Visible option.
3. Click Apply → Close.
4. Observe the radiation pattern displayed on the antenna structure.